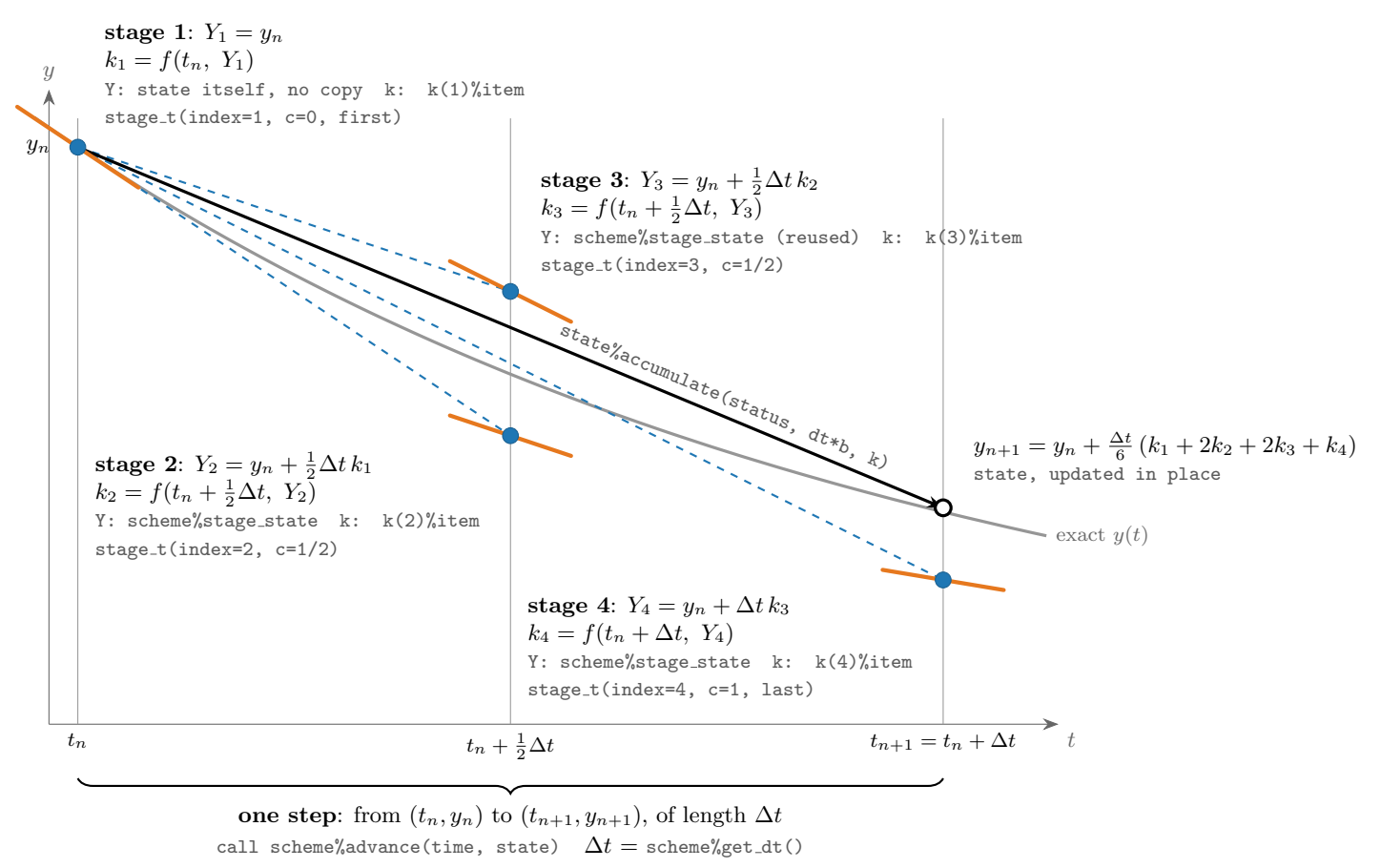


symbol	held in	produced by
y_n, y_{n+1}	state: the program's ODEmodel_state_c	updated in place at the end of the step: state%accumulate(status, dt*b, k)
• Y_i, stage state	Y_1: state itself, no copy; Y_2, Y_3, Y_4: scheme%stage_state, one buffer for all	stage_state%combine(status, state, dt*a(i,:), k), then model%apply_boundaries
— k_i, stage tendency	scheme%k(i)%item: an ODEmodel_tendencies_c, in a tendency_box_t	model%compute_tendencies(status, Y, t, scratch, k(i)%item, stage)
f	the model, an ODEmodel_c, with its scratch_space_c	written by you
i, c_i, Δt	stage_t: index, stages, c, dt, first, last	passed with every evaluation
c, a_{ij}, b	rk_tableau_t, the definition: it extends method_descriptor_t	rk_tableau("rk4"); constants in the hardcoded step
the step	a time_scheme_c, built by rk_scheme(scheme, "rk4", dt, model)	scheme%advance(time, state)

dashed: how a stage state is built, from y_n along the previous slope; →: the update, along the weighted average of the four slopes. Drawn exactly for $y' = -y$, $y_n = 1$, $\Delta t = 1$: the stage states are off the exact curve, $y_{n+1} = 0.375$ is on it ($e^{-1} = 0.368$).